PRACTICE SHEET

1. What is the equivalent definition of the function given by $f(x)=\begin{cases}2x, x \geq 0\\ 0, x<0\end{cases}$?

(a) $f(x) = |x|$ (b) $f(x) = 2x$
(c) $f(x) = |x| + x$ (d) $f(x) = 2|x|$

2. If $f : R \to R^+$ such that $f(x) = (1/3)^x$, then what is the value of $f^{-1}(x)$?

(a) $(1/3)^x$ (b) 3^x
(c) $\log_{1/3} x$ (d) $\log_x (1/3)$

3. Consider the following statements:
The function f (x) = greatest integer $\leq$ x, x $\in$ R is a continuous function
1.All trigonometric functions are continuous on R.
2.Which of the statements given above is /are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

4. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, then what is $f\left(\frac{2x}{1-x^2}\right)$ equal to?

(a) $(f(x))^2$ (b) 1
(c) 2f(x) (d) $f\left(\frac{1-x}{1+x}\right)$

10. Consider the function f:R$\to${0,1} such that:

$f(x)=\begin{cases}1 & \text{if x is rational}\\ 0 & \text{if x is irrational}\end{cases}$

Which one of the following is correct?
(a) The function is one-one into
(b) The function is many-one into
(c) The function is one-one onto
(d) The function is many-one onto

Directions (for next three): Each item under List I is associated with one or more items under List II.

List I (Function)
A. sin x
B. cos x
C. tan x

List II (Property)
(i) Periodic function (ii) Non-periodic function
(iii) Continuous at every point on $(-\infty, \infty)$
(iv) Discontinuous function
(v) Differentiable at every point on $(-\infty, \infty)$
(vi) Not differentiable at every point on $(-\infty, \infty)$
(vii) Has period π (viii) Has period 2π
(ix)Increases on $\left(0, \frac{\pi}{2}\right)$ (x)Decreases on $\left(0, \frac{\pi}{2}\right)$